

Project- Microsoft Sentinel

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Introduction

Microsoft Azure Sentinel is a cloud-native Security Information and Event Management (SIEM) and Security Orchestration, Automation, and Response (SOAR) solution. It's designed to help Security Operations Center (SOC) analysts detect, investigate, and respond to security threats across an enterprise's digital environment.

Step 1: Set up a free Azure Account

To create a free account, visit <https://azure.microsoft.com/en-gb/pricing/purchase-options/azure-account> and click the Sign-Up button in the Azure Free Account section. You will get a \$200 USD free trial for 30 days.



Start with an Azure free trial

Get \$200 free credit toward Azure products and services, plus 12 months of popular [free services](#).

[Start](#)

You will then be prompted to enter your username and password. If you do not have one, you will need to create one. You will then enter your profile details and add a phone number which will trigger a verification code to be sent to you. Add the code, then click next. You will need to add a credit or debit card for identity verification. However, you will not be charged if you decide to move to pay-as-you go pricing.



m.sharp03@outlook.com

Enter your password

Password

[Forgot your password?](#)

Next

[Other ways to sign in](#)

[Sign in with a different Microsoft account](#)

Create your Azure free account

-  Popular services free for 12 months
-  55+ services always free
-  USD200 credit to use in your first 30 days

No automatic charges

After your credit is over, we'll ask you if you want to continue with pay-as-you-go. If you do, you'll only pay if you use more than the free amounts of services.

Step 2: Create a Resource Group

A **resource group** is a container that holds related resources for an Azure solution. You will first need to create a resource group to organize your resources.

- Log in to the Azure Portal:**
 - Go to portal.azure.com and log in with your Azure account.
- Navigate to Resource Groups:**
 - In the left-hand menu, click on **Resource groups**.

Navigate



Click the blue button to create a resource group.

- Create a New Resource Group:**
 - Click on **+ Create** at the bottom.



- Subscription:** Select the subscription you want to use.

- **Resource Group Name:** Give your resource group a name, such as *SentinelResourceGroup*.
- **Region:** Select the region closest to your location or based on your organization's requirements.

Microsoft Azure

Search resources, services, and docs (G+)

Home > Resource groups >

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ Azure subscription 1

Resource group name * ⓘ SentinelResourceGroup

Region * ⓘ (Middle East) UAE North

Previous Next Review + create

- Click **Review + Create**, then **Create** after validation.

When you refresh the page, you should see the resource page that you just created.

Resource groups

Default Directory

+ Create ⚙️ Manage view ▾ ↻ Refresh ⬇️ Export to CSV 🔗 Open query | ⚙️ Assign tags

📘 You are viewing a new version of Browse experience. Some features may be missing. [Click here to access the old experience.](#)

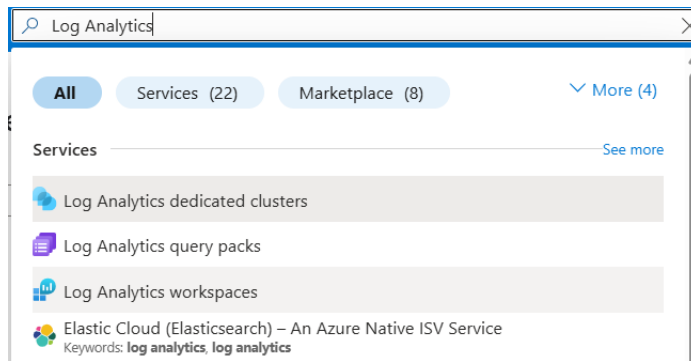
🔍 Filter for any field... Subscription equals all Location equals all × + Add filter

☐ Name ↑	Subscription	Location
☐ SentinelResourceGroup	Azure subscription 1	UAE North

Step 3: Create a Log Analytics Workspace

The **Log Analytics Workspace** is where Azure Sentinel will store its logs and data for analysis.

1. **Navigate to Log Analytics Workspaces:**
 - In the search bar at the top, type "Log Analytics" and select **Log Analytics Workspaces**.



2. Create a New Log Analytics Workspace:

- Click **+ Create**.



No log analytics workspaces to display

Leverage unique environments for log data from Azure Monitor and other Azure services, such as Microsoft Sentinel and Microsoft Defender for Cloud. Each workspace has its own data repository and configuration but might combine data from multiple services

[+ Create](#)

- **Subscription:** Choose your subscription.
- **Resource Group:** Select the resource group you created in Step 2.
- **Name:** Provide a name for your workspace, such as *SentinelLogsWorkspace*.
- **Region:** Choose the same region as your resource group.
- Click **Review + Create**, then **Create** after validation.

Create Log Analytics workspace ...

i A Log Analytics workspace is the basic management unit of Azure Monitor Logs. There are specific considerations you should take when creating a new Log Analytics workspace. [Learn more](#)

With Azure Monitor Logs you can easily store, retain, and query data collected from your monitored resources in Azure and other environments for valuable insights. A Log Analytics workspace is the logical storage unit where your log data is collected and stored.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Resource group * ⓘ [Create new](#)

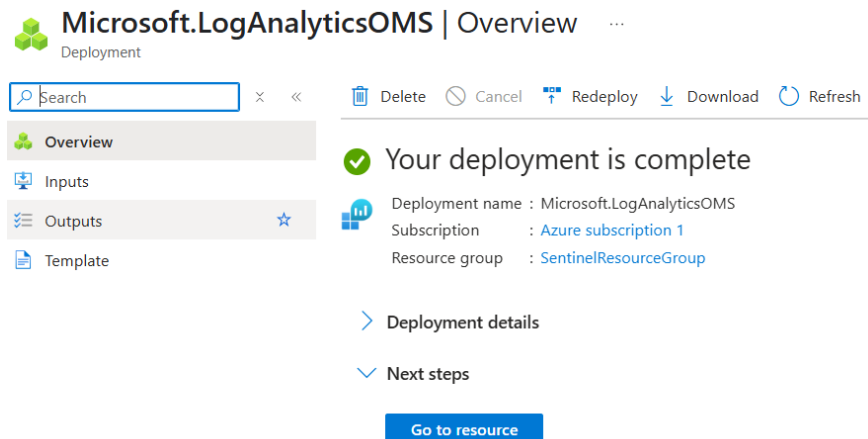
Instance details

Name * ⓘ ✓

Region * ⓘ

[Review + Create](#) [« Previous](#) [Next : Tags >](#)

After you have successfully deployed the workspace, the following notification will appear.

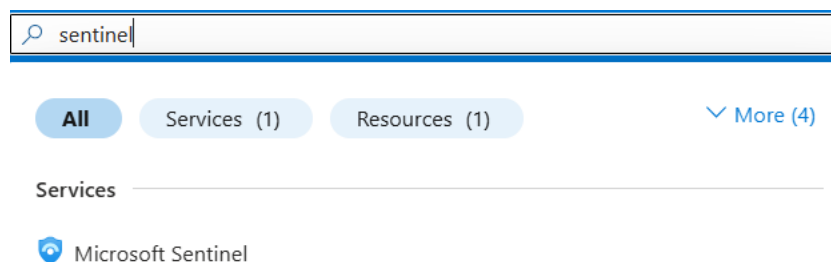


Step 4: Create Microsoft Sentinel

Microsoft Sentinel provides SIEM (Security Information and Event Management) capabilities, allowing you to monitor security across your infrastructure.

1. Navigate to Microsoft Sentinel:

- In the search bar, type "Microsoft Sentinel" and select it.



2. Add Microsoft Sentinel to Your Workspace:

- Click **+ Add** in the Microsoft Sentinel pane. Click the blue button to create a Microsoft Sentinel.



No Microsoft Sentinel to display

See and stop threats before they cause harm, with SIEM reinvented for a modern world. Microsoft Sentinel is your birds-eye view across the enterprise.



- **Select Workspace:** Choose the Log Analytics Workspace you created in Step 2 (SentinelLogsWorkspace).
- Click **Add** to attach Sentinel to your workspace.

Add Microsoft Sentinel to a workspace ...

+ Create a new workspace Refresh

Microsoft Sentinel offers a 31-day free trial. See [Microsoft Sentinel pricing](#) for more details.

Workspace ↑↓	Location ↑↓	ResourceGroup ↑↓	Subscription ↑↓	Directory ↑↓
SentinelLogsWorkspace	uaenorth	sentinelresourcegroup	Azure subscription 1	Default Directory

You will see it is added and activated with the free subscription trial for 30 days.

Microsoft Sentinel | News & guides ...

Selected workspace: 'sentinellogsworkspace'

Search Documentation

General

Overview

Logs

News & guides

Search

Threat management

Content management

Configuration

Microsoft Sentinel free trial activated

The free trial is active on this workspace from 5/10/2025 to 6/10/2025 at 11:59:59 PM UTC. During the trial, up to 10 GB/day are free for **both Microsoft Sentinel and Log Analytics**. Data beyond the 10 GB/day included quantity will be billed. [Learn more.](#)

OK

Collect and analyze data from any source, cloud or on-premises, in any format, at cloud scale. With AI on your side, find, investigate, and respond to real threats in minutes, with built-in knowledge and intelligence from decades of Microsoft security experience.

Step 5: Install the Windows Security Events Content Pack

Content packs provide pre-built templates, queries, and dashboards to help you monitor specific windows security events.

1. **Open Microsoft Sentinel:**
 - Go to **Microsoft Sentinel** from the portal.
2. **Add Data Connector:**
 - In the **Microsoft Sentinel** left panel, click on **Data connectors** from the left-hand menu. Next, scroll to the bottom of the page and click on the 'Go to content hub' option.

Microsoft Sentinel | N

Selected workspace: 'sentinellogsworkspace'

Search

General

Threat management

Content management

Configuration

Workspace manager (Preview)

Data connectors ☆

Analytics

Summary rules (Preview)

Watchlist

Automation

Settings

Getting started

Featured data connectors

These are the top data connectors for your data ingestion and onboarding needs.

Get these data connectors

More data connectors

Explore all connectors in content hub. Install on demand.

Go to content hub

In the search bar type in '**Windows Security Events**' and choose the **Windows Security Events** option. In the lower right corner of the page, click the blue install button.

Content title	Status	Content source	Provider	Support	Category	Content type
☒ Windows Security Events	☐ Not installed	Solution	Microsoft	Microsoft	Security - Threat Protection	Analytics rule (20) Data connector (2) +2
Windows Security Events via AMA	☐ Not installed	Solution	Microsoft	Microsoft	Security - Threat Protection	Data connector
Security Events via Legacy Agent	☐ Not installed	Solution	Microsoft	Microsoft	Security - Threat Protection	Data connector



Windows Security Events

Microsoft Provider

Microsoft Support

3.0.9 Version

The Windows Security Events solution for Microsoft Sentinel allows you to ingest Security events from your Windows machines using the Windows Agent into Microsoft Sentinel. This solution includes two (2) data connectors to help ingest the logs.

- Windows Security Events via AMA** - This data connector helps in ingesting Security Events logs into your Log Analytics Workspace using the new Azure Monitor Agent. Learn more about ingesting using the new Azure Monitor Agent [here](#). **Microsoft recommends using this Data Connector.**
- Security Events via Legacy Agent** - This data connector helps in ingesting Security Events logs into your Log Analytics Workspace using the legacy Log Analytics agent.

NOTE: Microsoft recommends installation of Windows Security Events via AMA Connector. Legacy connector uses the Log Analytics agent which is about to be deprecated by **Aug 31, 2024**, and thus should only be installed where AMA is not supported.

[Install](#)
[View details](#)

Refresh and go back to the data connectors page, then you will see the two connectors.

2 Connectors

0 Connected


[More content at Content Hub](#)

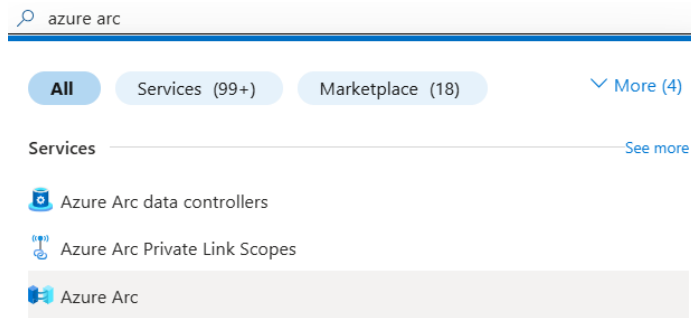
Providers : **Microsoft**

Status	Connector name ↑
	Security Events via Legacy Agent Microsoft
	Windows Security Events via AMA Microsoft

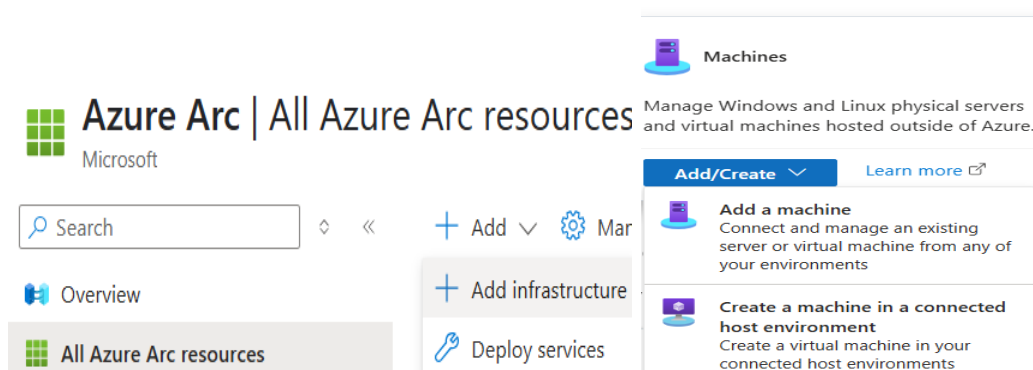
Step 6: Install Azure Arc on a Device

Azure Arc allows you to manage on-premise or multi-cloud resources in Azure.

- Navigate to Azure Arc:**
 - In the search bar, type **Azure Arc** and select **Azure Arc**.



In the left pane, choose 'All Azure Arc resources', followed by the add button, choosing 'Add infrastructure' from the dropdown menu. Click on the 'Add/Create' button in the Machines section and click 'Add a machine'.

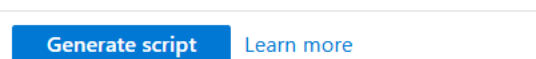


2. Add a Server:

- Click on **+ Add** in the Azure Arc pane.
- Choose **Servers** as the resource type you want to connect. Click on a "add a single server" window.

Add a single server

This option will generate a script to run on your target server. The script will prompt you for your Azure login, so this option is best for adding servers one at a time.



3. Generate the Script to Install Azure Arc:

- Choose your **Subscription** and **Resource Group** (*SentinelResourceGroup*).
- Provide a **Region** for the resource.
- **Download the script** provided by Azure Arc.

- Run the script on the target device (e.g., a Linux or Windows server) to install and register Azure Arc.

Project details

Select the subscription and resource group where you want the server to be managed within Azure.

Subscription * ⓘ

Resource group * ⓘ

[Create new](#)

Server details

Select details for the servers that you want to add. An agent package will be generated for the selected server type.

Region * ⓘ

Operating system * ⓘ

SQL Server

Connect SQL Server ⓘ ☐

ⓘ Automatically connect any SQL Server instances to Azure Arc. [Learn more](#) ⓘ

Connectivity method

Choose how the connected machine agent running in the server should connect to the Internet. This setting only applies to the Arc agent. Proxy settings for extensions are configured separately.

Connectivity method *

☒ Public endpoint
Connect to the internet directly using default routes on your server. This supports Arc Gateway.

☐ Private endpoint
Connect using an Azure virtual network private endpoint. Requires Express Route or a VPN. Arc Gateway is not supported.

Proxy server URL ⓘ

Gateway resource ⓘ

[Create new](#)

[Previous](#) [Next](#) [Download and run script](#)

On the next page download the created script to your device. Once downloaded, type in the 'powershell' in the search bar on your PC. To have all the required permissions, run Powershell as an administrator

Add a server with Azure Arc ...

[Download](#) 

2. Open a PowerShell console to run the script

Before you run the script, make sure your server meets the following requirements.

- HTTPS access to Azure services
The server requires access to port 443 and a set of outbound URLs for the Azure Arc agent to properly function. [View outbound URLs](#)
- Local administrator permission



Change your directories to the downloads folder using the command: `cd $HOME\Downloads`

```
PS C:\WINDOWS\system32> cd "$HOME\Downloads"
PS C:\Users\Mallikah\Downloads> cd $Home\Downloads
```

Run the script you have downloaded on your device. You may have to temporarily enable script execution.

Run the command: `Set-ExecutionPolicy RemoteSigned -Scope Process`

```
PS C:\Users\Mallikah\Downloads> Set-ExecutionPolicy RemoteSigned -Scope Process

Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing the execution policy might expose you to the security risks described in the about_Execution_Policies help topic at https://go.microsoft.com/fwlink/?LinkID=135170. Do you want to change the execution policy?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"):
```

You might be prompted to confirm if you trust this script. Type 'Y' and proceed using the command below.

Command: `.\OnboardingScript.ps1`

```
PS C:\Users\Mallikah\Downloads> .\OnboardingScript.ps1
VERBOSE: Installing Azure Connected Machine Agent
VERBOSE: PowerShell version: 5.1.26100.3624
VERBOSE: Total Physical Memory: 8192 MB
VERBOSE: .NET Framework version: 4.8.9032
VERBOSE: Checking if this is an Azure virtual machine
VERBOSE: Error Unable to connect to the remote server checking if we are in Azure
VERBOSE: Downloading agent package from https://gbl.his.arc.azure.com/azcmagent/latest/AzureConnectedMachineAgent.msi to C:\Users\Mallikah\AppData\Local\Temp\AzureConnectedMachineAgent.msi
VERBOSE: Installing agent package
```

You will be asked to confirm your identity by logging into your Microsoft Account.

Verify the Installation:

After running the script on your device, return to the Azure portal and verify that the device appears under the **All Azure Arc Resources** by refreshing the page.

+ Add Manage view Refresh Export to CSV Open query Assign tags

Filter for any field... Subscription equals all Type equals all Resource group equals all Location equals all Add filter

Showing 1 to 1 of 1 records. No grouping List view

Name	Resource group	Location	Subscription	Type
LAPTOP-VUMAL556	SentinelResourceGroup	UAE North	Azure subscription 1	Machine - Azure Arc

Step 7: Set Up a Data Collection Rule for the Azure Arc Device from Sentinel's Data Connector

Now that your device is connected via **Azure Arc**, you need to ensure the logs from that device are collected. You'll set up a **Data Collection Rule (DCR)** from the **Windows Security Events** Data Connector in Sentinel.

- Go Back to Microsoft Sentinel:**
 - Navigate back to **Microsoft Sentinel**.
- Access the Windows Security Events Data Connector:**
 - Click on **Data connectors** from the Sentinel menu.
 - Search for **Windows Security Events** via AMA (Azure Monitor Agent) click on the '**Open Connector Page**' button followed by the '+**Create data collection rule**'.

2 Connectors 0 Connected More content at Content Hub

Search by name or provider Providers : Microsoft Data Types

Status : Not connected (2)

Status	Connector name
	Security Events via Legacy Agent Microsoft
	Windows Security Events via AMA Microsoft

Windows Security Events via AMA

Disconne... Status	Micros... Provider	Last Log Re...
		--

Description

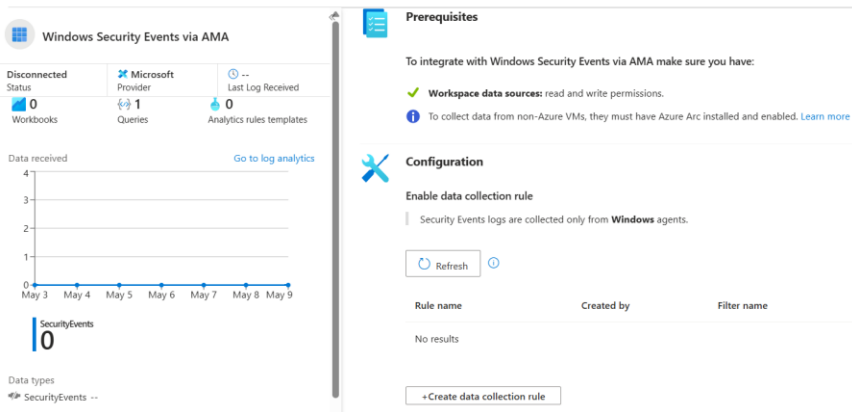
You can stream all security events from the Windows machines connected to your Microsoft Sentinel workspace using the Windows agent. This connection enables you to view dashboards, create custom alerts, and improve investigation. This gives you more insight into your organization's network and improves your security operation capabilities.

Last data received

--

Open connector page

Windows Security Events via AMA ...



3. Create Data Collection Rule (DCR):

- You'll be prompted to create a **Data Collection Rule (DCR)**.
- Provide a **Name** for your rule (e.g., `ArcSecurityDataCollection`).
- Select the **Subscription** and **Resource Group** (SentinelResourceGroup) where your Arc-connected device is located.
- Select the **Region** that matches your Arc-connected device.

Create Data Collection Rule

Data collection rule management

Basic Resources Collect Review + create

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all of your resources.

Rule details

Rule name *	ArcSecurityDataCollection
Subscription * ⓘ	Azure subscription 1
Resource group * ⓘ	SentinelResourceGroup

4. Add Data Sources:

- In the **Data Sources** section, select **Windows Security Events** and other relevant event types that you want to collect from your Arc-connected device.

5. Assign the Azure Arc-Connected Device:

- In the **Resources** section, select your **Azure Arc-connected device** from the list of available resources.

6. Select Log Analytics Workspace:

- Ensure the **Destination** of the logs is set to your **Log Analytics Workspace** (`SentinelLogsWorkspace`).

7. Review and Create:

- Review your settings, then click **Next: Collect** to finalize the Data Collection Rule.

Create Data Collection Rule

Data collection rule management



Choose a set of machines to collect data from. This set of machines will replace any previous selection, make sure to re-select any you'd like to keep. The Azure Monitor Agent will automatically be installed.

- This will also enable System Assigned Managed Identity on these machines, in addition to existing User Assigned Identities (if any). Note: Unless specified in the request, the machine will default to using System Assigned Identity for all other applications. [Learn more](#)

Subscriptions	Resource Groups	Resource Types	Locations
Selected: All	Selected: All	Selected: All	Selected: All

Scope	Resource Type	Location
☒ Azure subscription 1		
☒ sentinelresourcegroup		
☒ LAPTOP-VUMAL556	microsoft.hybridcompute/machines	UAE North

< Previous

Next: Collect >

Leave the collect rule management the same as "All security events".

Create Data Collection Rule

Data collection rule management

Basic Resources **Collect** Review + create

Select which events to stream. ⓘ

☒ All Security Events ☐ Common ☐ Minimal ☐ Custom

Then review and create your Data Collection Rule. You will see a confirmation of your rule creation in the Windows Security Events via AMA connector page.



Prerequisites

To integrate with Windows Security Events via AMA make sure you have:

- ✔ **Workspace data sources:** read and write permissions.
- ℹ To collect data from non-Azure VMs, they must have Azure Arc installed and enabled. [Learn more](#)



Configuration

Enable data collection rule

Security Events logs are collected only from **Windows** agents.

ⓘ

Rule name	Created by	Filter name
ArcSecurityDataCollection	Sentinel	AllEvents Edit Delete

You have successfully set up the Microsoft Sentinel collecting data from your device.

Conclusion

A Security Information and Event Management (SIEM) system is essential to the work of a SOC analyst. It empowers analysts to detect, investigate, and respond to security incidents with greater speed and accuracy. With features like automated alerts, customizable dashboards, and advanced analytics, SIEMs help pinpoint suspicious behaviour and uncover potential vulnerabilities. By automating routine tasks, they improve operational efficiency and free analysts to focus on critical threats. In today's fast-changing threat landscape, where attacks can emerge and evolve quickly, a SIEM is crucial to ensuring no threat goes unnoticed.